Lab 4: Vibe Coding - Chat Transcript (Endless Runner)

Candidate SRN: PES1UG24AM105 | **Assigned Project:** Real-Time Endless Runner Game (39_endless-runner)

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LAB 4: VIBE CODING - PAIR PROGRAMMING CHAT TRANSCRIPT
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Candidate SRN       : PES1UG24AM105
Assigned Project    : Real-Time Endless Runner Game (39_endless-runner)
Course / Module     : Software Engineering Lab 4 (Vibe Coding)
Pair Programmer     : Google DeepMind Gemini Assistant
Deliverable         : Deliverable (c) - Complete LLM Chat History
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[TURN 1] USER (PES1UG24AM105)
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Lab 4: Vibe Coding
Objective:
Use Vibe Coding tools to fix the broken code and add features to the same.
Learn how to prompt to get the job done in under three to four attempts.
Clone a repo assigned to you, make changes and push to your repo in your
personal account.
THIS IS AN INDIVIDUAL ASSIGNMENT. WORK INDIVIDUALLY!!

I have downloaded the repo into C:\LAB4_105.
My assigned repo is 39_endless-runner-main and my SRN is: PES1UG24AM105.

From the README.md:
What's Provided:
- A player-controlled character that jumps over obstacles with gravity
pulling it back down
- Obstacles that spawn at a regular interval and scroll toward the player,
gradually speeding up
- Score display

Tasks to Complete:
Task 1: Refine Collision Detection
The game speeds up forever with no limit, and once it's fast enough
obstacles can zip past the player without the hit ever registering.
Investigate and enhance collision accuracy (and/or the speed ramp) so it
stays fair no matter how long a run lasts.
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Task 2: Implement Game Over Condition
Add a screen that displays the final score once the player collides with an obstacle, then gracefully waits for input instead of just printing to the console.

Task 3: Add Replay Option
After Game Over, allow the user to play again by choosing a difficulty (Easy, Medium, or Hard starting speed/spawn rate), or exit.

Task 4: Add Sound Feedback
Add basic sound effects for jumping, passing an obstacle (scoring), and the game-over moment.

Additional Features Requested:
1. Start Screen: When running the game, it opens into an attractive Start Screen with "START GAME" button (or Space/Enter) and difficulty selection.
2. Maximize Option: Allow the window to be maximized with proper scaling.
3. Candidate identification: Include my SRN PES1UG24AM105 in the game.

Please complete this in the best way possible.

[TURN 2] AI PAIR PROGRAMMER

Let's analyze the Endless Runner codebase for SRN: PES1UG24AM105 and solve all required tasks.

1. Root Cause Analysis of Bugs:
* Task 1 (High-Speed Tunneling Bug):
  In the starter game_engine.py:
    self.speed += self.speed_increase_per_frame # 0.003 every frame with NO ceiling
  Obstacle movement: obstacle.x -= self.speed
  Collision is checked only after the movement:
    if obstacle.rect().colliderect(self.player.rect()):
  As speed increases, the horizontal displacement covered in one frame exceeds the player's 30px width. An obstacle can be to the right of the player on frame N and entirely to the left on frame N+1, skipping the hitbox entirely ("tunneling").
  Fix:
  1. Continuous Collision Detection (CCD) via Swept-Box: In obstacle.py, record `prev_x`. The swept path during the frame spans [min(x, prev_x), max(x + w, prev_x + w)]. If this displacement intersects the player's horizontal position while vertical clearance overlaps, collision is reliably detected.
  2. Speed Bounds: Implement difficulty-specific speed ceilings (`max_speed = 11.0` for Easy, `14.0` for Medium, `16.5` for Hard) with tuned acceleration so high-speed runs stay challenging yet physically fair.

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* Task 2 (Game Over Condition):
  The original code merely executed `print("Game over! Final score:",
self.score)` and froze the Pygame window.
  Fix: Render an in-game modal card overlay displaying "GAME OVER", Final
Score, Best Score, and Candidate SRN: PES1UG24AM105.

* Task 3 (Replay & Difficulty Selection):
  Implemented in-place clean state reset supporting:
  - [1] Easy: Starting speed 5.0, spawn interval 85 frames, max speed 11.0
  - [2] Medium: Starting speed 7.0, spawn interval 65 frames, max speed 14.0
  - [3] Hard: Starting speed 9.0, spawn interval 50 frames, max speed 16.5
  - [Space / R]: Quick restart
  - [Q / Esc]: Exit

* Task 4 (Sound Feedback):
  Generated standalone 16-bit PCM WAV sound effects via standard library
`wave` and `math`:
  - jump.wav: Upward pitch sweep on jumping
  - score.wav: Pleasant two-tone chime when clearing an obstacle
  - hit.wav: Low impact crash thud on obstacle collision
  Managed by `SoundManager` with automatic fallback handling.

* Additional Requested Enhancements:
  - Start Screen: Interactive Title Screen with animated runner, difficulty
selection buttons, and prominent `[ START GAME ]` button.
  - Screen Maximize & Resizing: Configured with `pygame.RESIZABLE`, aspect-
ratio preserving scaling (800x400) via `pygame.transform.smoothscale`, and
virtual mouse coordinate translation.
  - Candidate Identification: Window title set to "Endless Runner -
PES1UG24AM105", HUD and Game Over screen displaying SRN: PES1UG24AM105.

### 2. How to Record the First Video (Before Changes):
Run:
  cd C:\LAB4_105\endless-runner-original
  python main.py
Demonstrate in 10 seconds:
1. Game drops straight into play with no start screen and total silence.
2. Obstacles accelerating infinitely.
3. Sudden freeze on collision with no Game Over screen (logs only to
console).

### 3. How to Record the Second Video (After Changes):
Run:
  cd C:\LAB4_105\endless-runner-updated
  python main.py
Demonstrate in 10 seconds:
1. Start Screen with "ENDLESS RUNNER", Candidate SRN: PES1UG24AM105, and
START GAME button.
2. Screen Maximize button on the title bar showing smooth letterboxed
scaling.
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3. Sound effects on jumping, clearing obstacles, and collision.
4. Continuous collision detection registering hits without tunneling.
5. In-game Game Over screen and difficulty replay ([1], [2], [3]).

[TURN 3] USER (PES1UG24AM105)

How to run these same files on a MacBook?

[TURN 4] AI PAIR PROGRAMMER

Here is how to run the Endless Runner project on macOS:

1. Copy the folder to your Mac (e.g. to Desktop).
2. Open Terminal (Cmd + Space -> Terminal).
3. Verify Python 3:
     python3 --version
4. Create a virtual environment and install dependencies:
     cd ~/Desktop/endless-runner-updated
     python3 -m venv venv
     source venv/bin/activate
     pip install pygame
5. Run the game:
     python3 main.py
6. Screen Recording on Mac:
   - Press Cmd + Shift + 5 to open the native screen recorder.
   - Select the game window and record 10 seconds.
   - Video automatically saves to the Desktop as a .mov file.
7. Mac Controls:
   - Click the green window button to zoom/maximize.
   - Press Fn + F11 to toggle fullscreen.

[TURN 5] USER (PES1UG24AM105)
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c. Chat history exported as a doc/pdf - this is one of the deliverables that I found. How to do it?

[TURN 6] AI PAIR PROGRAMMER

The complete chat history has been formatted and exported specifically for

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SRN: PES1UG24AM105:
- PDF Deliverable: Lab4_Chat_History_PES1UG24AM105.pdf
- Document / Text Deliverable: Chat_History_EndlessRunner_PES1UG24AM105.txt

Manual Export Instructions:
1. In your browser or chat tool, press Ctrl + P (or Cmd + P on Mac).
2. Set Destination to "Save as PDF".
3. Check "Background graphics" under More Settings.
4. Click Save to your lab submission directory.
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END OF TRANSCRIPT - ENDLESS RUNNER (PES1UG24AM105)
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